Fiscal Year:	FY 2018	**Task Last Updated:**	FY 11/14/2018
PI Name:	Kondic, Lou Ph.D.		
Project Title:	Structure Evolution during Phase Separation in Colloids under Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	07102-1824	**Congressional District:**	10
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-B: Use of the NASA Physical Sciences Informatics System – Appendix B
Start Date:	08/16/2016	**End Date:**	08/15/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Khusid, Boris Ph.D. (New Jersey Institute of Technology) Meyer, William Ph.D. (Universities Space Research Association)		
Grant/Contract No.:	NNX16AQ79G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Motivation: Numerous studies of phase transformations in colloids are motivated not only by scientific curiosity but also by myriad technologies in materials engineering, electronics, life science, and chemical and pharmaceutical industries. A number of theories for phase transition in colloids have been proposed in the literature. As structure transformation in colloids in terrestrial experiments is influenced by detrimental gravity effects, such as buoyancy-driven convection and particle sedimentation, using Earth-based studies for testing colloidal theories is problematic. To uncover mechanisms governing phase transformations, a series of microgravity experiments was carried out over the last several decades on Space Shuttle to study colloids without masking gravity effects. Objectives: The main goal of the proposed research is to compare the results of the Space Shuttle experiments with the ones carried out in terrestrial environment by developing accurate models and simulations of predictive power. Methodology: Models for structure transformations in colloids undergoing phase separation have been developed and the results compared directly to the experimental ones. Deliverables: Better understanding of phase transformations in colloidal systems, analyzed microscopy images recorded in Shuttle experiments, phase-field models for phase transformations in colloids, and codes for numerical computations of the developed models. New models will provide a benchmark for comparison and evaluation of existing colloidal theories that were developed based on terrestrial experiments influenced by gravity-driven processes. Effective computational modeling will help improve Earth-based production of advanced materials. The research will be integrated with educational and outreach initiatives in space sciences and related technologies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Effective computational modeling will help improve Earth-based production of advanced materials.
Task Progress:	The project focused on development of a model describing growth of colloid nuclei. The developed model has been tested by direct comparison with the experimental results carried out on Space Shuttle, as well as with the terrestrial experiments. The model is based on the classical approach; however, it includes additional information and findings, in particular regarding the equation of state, which have become available only relatively recently. The model includes modeling of coupled diffusion problems for the solid and liquid phase, combined with the consistent boundary conditions involving osmotic pressure balance including interfacial tension. The motion of the solidification front is modeled based on Wilson-Frenkel law. In parallel with developing the model, we have also developed novel computational scheme that allows for accurate and efficient simulations. This scheme, formulated for radially symmetric problems, is of finite-volume type and it belongs to front-capturing type of schemes. Such a scheme allows for precise and accurate treatment of the boundary conditions at fluid/solid boundary including accurate conservation of mass. The main outcome of the project is significant new insight into the mechanism of colloidal nuclei growth, showing in particular that implementation of nonlinear diffusion is crucial for proper understanding of the problem. In addition, by comparing directly the results of Space Shuttle experiments with the ones obtained under Earth gravity, we have been able to show consistency between the considered experiments. This finding provides new insight into formation of crystalline and glassy structures in the considered experiments. A manuscript, "Growth of colloid nuclei," by M. Lam, B. Khusid, W. Meyer, L. Kondic, is in process to be submitted a a peer-reviewed journal.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Abstracts for Journals and Proceedings	Kondic L, Lam M, Khusid B, Meyer W. "Growth and Interaction of Colloid Nuclei under Microgravity." German Physical Society (DPG) Annual Meeting, Berlin, Germany, March 2018. Proceedings of DPG. German Physical Society (DPG) Annual Meeting, Berlin, Germany, March 2018. , Mar-2018
Abstracts for Journals and Proceedings	Lam M, Khusid B, Meyer W, Kondic L. "Growth and Interaction of Colloid Nuclei." 70th Annual Meeting of the APS Division of Fluid Dynamics, Denver, CO, November 19-21, 2017. Bulletin of the American Physical Society. 2017 Nov;62(14):BAPS.2017.DFD.A35.5. http://meetings.aps.org/Meeting/DFD17/Session/A35.5 , Nov-2017
Abstracts for Journals and Proceedings	Kondic L, Lam M, Khusid B, Meyer W. "Growth and Interaction of Colloid Nuclei." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Kondic L, Lam M, Khusid B, Meyer W. "Modeling Growth of Colloid Nuclei." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Articles in Peer-reviewed Journals	Lam MA, Khusid B, Kondic L, Meyer WV. "Role of diffusion in crystallization of hard-sphere colloids." Physical Review E. 2021 Nov;104(5):054607. https://doi.org/10.1103/PhysRevE.104.054607 , Nov-2021